

Fish Size Detection and Classification

Using Computer Vision and Machine Learning Techniques

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Overview

- **Objective:** Automate the detection and classification of fish sizes.
- **Approach:** Various techniques including scale detection, labeled sheets detection, and segmentation for size prediction.

Motivation and Challenges

Importance: Accurate fish size classification is vital for fisheries and research.

Challenges:

- Varied fish positioning from the camera.
- Difficulty in capturing the scale along with three fish in an image.

Methods Overview

1. **Initial attempts**
2. **Scale Detection Model**
3. **Size Prediction Without Scale**
 - Using "Good" and "Bad" labeled cards.
 - Yellow Graph Sheet Approaches.
4. **Manual Size Labeling and Regression Models**

Initial Attempts

1. Contour-Based Scale Detection:

- Initial attempts to extract the scale part from the image using contours faced challenges, as the scale was not detected accurately. There were uncertainties whether the detected object was the scale or another element in the image.

2. ML Regression and Classification Models:

- Trained ML regression models using segmented fish image pixels to predict length. However, the mean squared error rate was high (**1.15**), indicating inaccurate predictions. Total data used was 588 images.

- Attempted ML classification models but encountered bias due to imbalanced training data, particularly towards the 17cm class.

Scale Detection Model

Goal: Measure fish dimensions in centimeters using a scale detected in the image.

Dataset: Trained YOLO segmentation model on 753 images with a yellow 15 cm scale. By isolating the measurement-specific part of the scale, the model can effectively function with any scale image, providing greater versatility and accuracy. 551 train, 151 validation and 51 test images.

Challenges: Capturing the entire scale along with 3 fish in the image.

Size Prediction Without Scale

- **Using "Good" and "Bad" Cards:**
 - Labeled rectangular cards to estimate fish dimensions.
 - Cards' width was larger than the average Sardine width (5 cm vs. 2.6 cm).
- **Issues:** Difficulty fitting cards and fish together in images.

Yellow Graph Sheets Approach

Hough Transform Method: Attempted line detection but faced inconsistency.

YOLO Segmentation:

- Model trained with labeled squares, but it failed to generalize.

YOLO Detection with Matched RGB Values:

- Performed well on synthetic data but grouped squares in real images.

Regression Models for Fish Size Prediction

- Manually annotated 830 images, used 812 for training and validation.
- Feature Selection: Removed low correlation features like image height, width.
- Features converted to centimeters using known fish data.
- Best Regression model (AdaBoost) had an MAE of 1.1 and a Max Absolute Error of 3.5.

Size Based freshness classification

1. Extract scale size in pixels using the YOLO segmentation model trained to detect scales in the image.
2. We set the reference standard scale length to 16cm as we are taking images with 15 cm scales in them along with the fish.
3. Calculated fish height, width and other dimensions using the detected scale length as a reference to convert pixels to centimeters for precise measurement.
4. $\text{Centimeter_to_pixel} = 15\text{cm}/\text{scale_length_in_pixels}$
5. Utilized contour detection to calculate fish circumference and area, converting them to centimeters for accurate assessment.

EDA

- Conducted EDA on 697 bad and 697 good images, analyzing features such as length, circumference, and area. Observed differences between the two classes, with the minimum length of the good class consistently around 13cm.
- This is the complete dataset of fish labeled as bad due to “size” and fish labeled as “good” on the same dates.

Feature Extraction

We have meticulously engineered a set of features to train our ML models. These features encompass various aspects of the fish's dimensions and the scale present in the image.

One_Pixel_Value: A crucial metric representing the number of centimeters per pixel.

Circumference(cm): The circumference of the fish.

Area(cm): The area occupied by the fish

FishHeight(cm), FishWidth(cm), FishDiagonal(cm)

ScaleDiagonal(cm), ScaleWidth(cm)

Machine Learning Models

1. Logistic Regression
2. Decision Tree
3. Random Forest Classifier
4. Support Vector Classifier
5. XGBoost
6. AdaBoost
7. Gradient Boost
8. K Nearest Neighbours
9. Extra Trees
10. Naive Bayes

ML Model for Classification

1. Created a dataframe with features extracted from EDA for training ML models to classify images into good and bad classes.
2. ML classification models were trained with various feature combinations, including fish dimensions, scale data, and cms_per_pixel.
3. we balanced the dataset by selecting 160 bad and 160 good images for training, with an additional 40 images from each class for validation.
4. Logistic Regression emerged as the top performer according to the train and validation accuracies.
5. Train accuracy: 98%, validation accuracy: 95%, test accuracy: 94% with threshold 0.5

Threshold-Based Size Classification

Previous Model: Logistic regression model's predictions with 0.5 threshold, whose input features are different fish dimensions converted to centimeters.

Condition: If the diagonal length < 12 cm, classify as "Bad."

Rationale: Diagonal length accounts for fish positioning in the image.

Recent Size Based ML Classification Model Input Features

- FishWidth(cm), FishDiagonal(cm)
- ScaleDiagonal(cm), ScaleWidth(cm)
- OnePixelValue, AspectRatio
- DiagonalToHeightRatio, Area Ratio
- Circumference(cm), Area(cm)

Exploratory Data Analysis (EDA)

- Scatter plots and histograms for each feature were plotted, but no significant insights emerged.
- This is likely due to the diverse rejection reasons for the Bad class, not limited to size alone.

Results

Experiment	Dataset	Train data (per class)	Test data (per class)	Unseen data (per class)	Model	Test accuracy	Unseen accuracy	Test Bad f1-score	Test Good f1-score
1	data collection only	1467	368	100	SVC	93%	92%	93%	93%
					XGB	92%	94%	92%	92%
2	App testing only	162	40	50	RandomForest	77%	74%	76%	77%
					Gradient Boosting	77%	75%	76%	77%
3	combined	2086	522	100	RandomForest	92%	90%	92%	92%
					XGB	92%	93%	91%	92%

With increased dataset

Incorporated data collected till **December 26, 2024** along with the app testing data till **December 22, 2024**.

Dataset Overview:

- **Total Data Points:** 4299 Good | 2836 Bad
 - **Data Collection Data:** 2258 Good | 1937 Bad
 - **App Testing Data:** 1321 Good | 666 Bad
 - **December Data:** 720 Good | 233 Bad

With increased dataset

- **Train/Validation/Unseen Split:**
 - Training: 2188 per class
 - Validation: 548 per class
 - Unseen: 100 per class
- **Additional Unseen Dataset 2:** 1463 Good | 10 Bad

With increased dataset

XGB Results:

Dataset	Accuracy
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Validation	82%
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Unseen 1	81%
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Unseen 2	85%
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- **December Data Only: 86%**
- **App Testing (Aug–Dec): 85%**
- **App Testing (Till Aug): 85%**

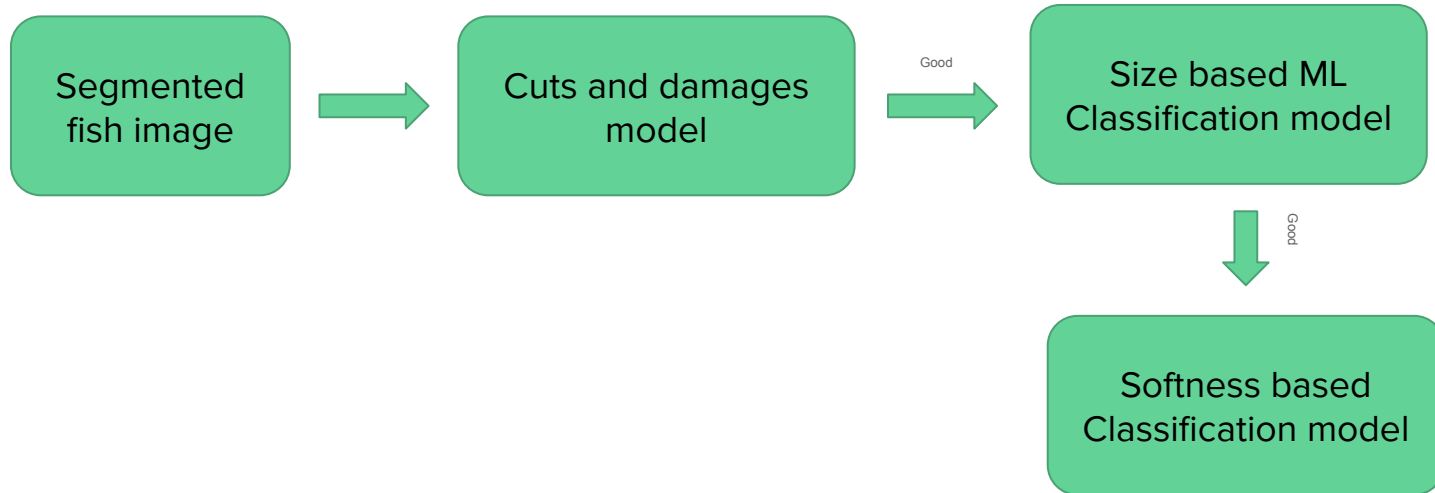
Sardine Model Workflow

1. Cuts and Damages Model: The image is first passed through the cuts and damages model. If the model predicts a bad outcome, the process concludes with a 'bad' classification.

2. Size Model: If the cuts and damages model predicts good, the image is then routed to the size model. Here, the model evaluates the size of the sardine based on a predefined scale height of 15 centimeters. If the size based ML classification model predicts bad, the process concludes with a 'bad' classification.

Sardine Model Workflow

3. Softness Model: If the size model predicts good, the image proceeds to the softness based classifying CNN model. This model assesses the softness of the sardine and provides its predictions.



Challenges and Lessons Learned

Challenges:

- Difficulty detecting squares consistently in graph sheets.
- Time-consuming manual labeling of data.

Lessons Learned:

- Manual labeling can create high-quality datasets.
- Alternative size prediction methods may be needed.

Future Work

Hyperparameter Tuning: Implement Grid Search CV for optimization.

Expand Dataset: Include additional labeled data and scale types.

New Techniques: Explore deep learning CNN models for more accurate predictions.

Conclusion

Key Achievements: Automated fish size classification with 96% accuracy with scale.

Remaining Issues: Some methods did not generalize well to real-world data.

Next Steps: Continue refining models and expanding dataset.

Thank You!